

Course Syllabus**GRAD-E1508: Evaluating AI Systems: Designing Experiments for Human-AI Performance**

Please note: This syllabus is a working draft. The dates and deadlines for assignments and readings are not yet finalized and may change. Confirmed dates will be announced before the start of the semester and posted on the course GitHub.

1. General information

Course Format	Onsite
Instructor(s)	Drew Dimmery
Instructor's E-mail	d.dimmery@hertie-school.org
Assistant (if applicable)	Ubaida Ansari
Instructor's Office Hours	Drop-in hours announced on first class, or contact professor for an appointment

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#).

For information on **course room, times and session dates**, please consult the Course Schedule on *MyStudies*.

Instructor Information:

Drew Dimmery is Professor of Data Science for the Common Good at the Hertie School. His research interests include causal inference, machine learning and the usage of data science for public policy. Prior to moving back to academia, he was a research scientist at Facebook in the United States for four years. He holds a PhD in political methodology from New York University and has published in top machine learning journals such as ICML, KDD and AISTATS and general interest journals like Science, Nature and Proceedings of the National Academy of Sciences.

2. Course Contents and Learning Objectives**Course contents:**

This course teaches students to design, pre-register, execute and describe rigorous evaluations of AI systems, drawing on the methodology of experimentation and measurement theory from the social sciences. AI systems are tools humans use to accomplish tasks; evaluating them is therefore fundamentally a question about joint human-AI performance on real tasks, not about the abstract model capabilities that static benchmarks measure. The course is structured around team projects carried through the full arc of design and execution: roughly the first two-thirds of the term is spent designing and pre-registering an evaluation, and the last third is spent running it, dealing with the inevitable surprises, and writing it up. Teams choose between two design archetypes (a human-subjects experiment with some form of AI as the intervention, or a model-only evaluation grounded in a real human task) and pursue a question of their own choosing. Teams are provided resources for accessing AI APIs and real human subjects. Students leave with one executed evaluation written up as a short technical report and a working understanding of the methodological landscape of AI evaluation.

Learning objectives:

Students will:

- Understand why AI evaluation is fundamentally a measurement challenge, and be able to apply social-science measurement frameworks (construct validity, operationalization) to AI systems.
- Be able to design, pre-register, and execute an experiment to evaluate whether an AI system supports humans in accomplishing a specific task.
- Be able to critically read and assess existing AI benchmarks, leaderboards, audit reports, and evaluation studies in both academic and industry contexts.
- Have hands-on experience with at least one open-source evaluation framework, and with running evaluations against both open and closed-weight models.
- Understand the common threats to validity in AI evaluation (benchmark contamination, prompt sensitivity, elicitation effort, model drift, over-reliance) and how to defend against them.
- Know how to report evaluation results honestly, including what went wrong, what had to be changed from the pre-analysis plan, and what the results do and don't support.

Prerequisites:

Students should have completed a causal inference course (or equivalent) and be comfortable writing code in Python or R for statistical analysis. No prior experience with LLMs is required, but students should be willing to use AI systems regularly throughout the course.

Diversity Statement:

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible.

3. Grading and Assignments

Policy on the use of AI tools for assignments:

The use of generative AI tools (such as ChatGPT) is **permitted throughout this course (except for closed book in-class assessments)**. Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to indicate clearly how they were used and to ensure the originality and academic integrity of their work. **Students should include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School: Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

Note that the *subject* of this course is the rigorous evaluation of AI systems. Students are expected to use AI tools throughout the term and to bring critical attention to how those tools shape their own work.

Composition of Final Grade:

Assignment	Deadline	Format	Weight
Assignment 1: Reading reflection memos (×2 per student)	Throughout term, due before the relevant session	Submit via Github	20%

Assignment 2: Ethics and responsible-evaluation self-assessment	Before session 6	Team submission (ethics steward)	10%
Assignment 3: Pre-analysis plan	End of week 8	Team submission (PAP steward), registered to OSF	20%
Assignment 4: Implementation work-in-progress presentations	Sessions 9–10	Team presentation (implementation steward)	10%
Assignment 5: Final team report	End of standard exam period	Team submission (report steward) via Github	30%
Participation	—	—	10%

Assignment Details:

The team deliverables below are each coordinated by a named member. At the start of the project, each four-person team assigns one member to each of four roles: an **ethics steward**, a **pre-analysis-plan steward**, an **implementation steward**, and a **report steward**. Each steward is responsible for the organization and coordination of one graded deliverable — the ethics self-assessment (Assignment 2), the pre-analysis plan (Assignment 3), the implementation work-in-progress presentations (Assignment 4), and the final report (Assignment 5), respectively — and **each of those four deliverables should be owned by a different member**. Stewardship is a managerial role rather than a heavier workload: a steward shifts their own effort on that deliverable toward *organizing* it — breaking the work into issues and distributing it, making sure each piece gets done when it needs to, reviewing teammates' pull requests, and keeping the deliverable's big picture in view — and correspondingly away from doing the bulk of the hands-on implementation (such as writing the code) themselves. This should show up in the GitHub history as organizational work — issues created and assigned, reviews, and project-board management — rather than as a larger share of the commits. The implementation steward also makes sure the experiment itself runs: that the evaluation pipeline gets built and the reproducible code-and-data repository that accompanies the final report stays runnable, coordinating and reviewing that work rather than writing all of it. The team still does the substantive work together; each steward owns coordination, final assembly, and accountability for their piece. The intent is to distribute clear ownership across the arc of the project so that no single member carries (or coasts on) the whole thing. On each of Assignments 2–5, 5 of the assignment's points — each point is one percentage point of the final grade — are set aside for **process**: the quality of the working process the team uses, described under *Working in the open on GitHub* below. This is assessed from the team's GitHub history and each student's project reflection.

These roles distribute coordination, not responsibility: every deliverable remains the whole team's collective responsibility, and the team is accountable for all four being completed. A full team is four people, one steward per deliverable; a smaller team (for example, three members after someone withdraws or falls ill) still assigns a steward to every deliverable, with one member stewarding two if necessary, and still completes all four. A smaller team may narrow the scope of its study to match its capacity, but no deliverable may be dropped.

Assignment 1: Reading reflection memos

Each student writes **two** short reflection memos over the term, each covering one week's required reading. Memos are at most one page and are submitted on Github 24 hours before the corresponding session.

The prompt is project-oriented: *what does this reading tell you about your team's evaluation? What insights or lessons does it offer? What, if anything, would you change about your design or analysis as a result?* This is a project-focused memo, not a referee report or a paper summary; the goal is to help you read each week's material with your project in mind, and to seed conversation in class.

Within each four-person team, students must coordinate so that **memos do not overlap**: each student covers different sessions than their teammates. Across the term, this means each team collectively writes eight memos that span a wide array of sessions. Teams will report their assignments in the appropriate part of the Github organization.

These memos will be graded by completion.

Assignment 2: Ethics and responsible-evaluation self-assessment

Before session 6, each team completes a responsible-evaluation self-assessment applied to its own design, owned by the ethics steward. Every team does this; which sections apply depends on the design archetype, and a team completes whichever genuinely bear on its study. Evaluating an AI system is never ethics-free: a model-only evaluation simply raises a different set of concerns than a human-subjects experiment.

Teams running a **human-subjects experiment** complete the Hertie School's **Master Programmes Self-Assessment Checklist** for research ethics and data protection: informed consent, vulnerable participants, deception or covert observation, sensitive topics and risk, and, on the data side, what personal data is collected and how it is kept confidential, reused, stored, and transferred under the GDPR. This feeds the ethics-review process arranged through Hertie.

Teams running a **model-only evaluation/benchmark** instead produce a short **evaluation data and responsibility statement**, in the lineage of [datasheets for datasets](#) (Gebru et al., 2021) and [model cards](#) (Mitchell et al., 2019). It covers: the provenance, licensing, and terms-of-service compliance of the evaluation data and the model APIs, and any personal data the data contains; the labor conditions and potential exposure to harmful content of any human raters or crowdworkers used to score or validate outputs; dual-use and harmful-capability concerns where the evaluation elicits unsafe behaviour; responsible disclosure of any harmful behaviour or vulnerability found in a deployed system; and honest, non-misleading comparison across systems. A model-only team that recruits human raters, or that handles personal or scraped data, also completes the corresponding sections of the Hertie checklist; the two tracks are not mutually exclusive.

Whichever track a team follows, the assessment forces it to reason explicitly, before committing to a design, about the ethics of its own evaluation. It feeds directly into the pre-analysis plan and is graded on the seriousness and specificity with which the team engages the concerns its own design actually raises.

Grading (10 points): 5 for the self-assessment itself and 5 for process.

Assignment 3: Pre-analysis plan

Each team submits a single, complete pre-analysis plan (PAP) at the end of week 8 and registers it to OSF (or an equivalent public registry). The PAP must specify the research question, construct, instrument, outcome measurement and scoring strategy, randomization and sampling design, primary and secondary hypotheses, estimation procedure, planned covariate adjustment, handling of missing data and non-compliance, and the team's defenses against the threats to validity discussed in class. A literate, executable PAP (notebook or equivalent) is preferred but not required.

From the moment the PAP is registered, **deviations from the plan must be documented and justified in the final report**. The PAP itself is graded on the rigor and credibility of the proposed design, on the quality of the team's engagement with threats to validity, and on the clarity and reproducibility of the document.

Grading (20 points): 5 for process, 5 for the overall research design, 5 for the analysis plan, and 5 for the constructs.

Assignment 4: Implementation work-in-progress presentations

During the implementation phase, each team gives two work-in-progress presentations in class — at session 9 (pilot results) and session 10 (mid-course status) — owned and run by the implementation steward. These are deliberately **informal**: a code-demo and status report, not a polished talk (they should not require preparation beyond the implementation that is already underway). Each should show the current state of the evaluation pipeline, the first interim or pilot results, what is breaking or surprising, and the specific questions on which the team most wants feedback. The bar is *prepared and specific enough to draw out useful feedback, and no more than that*. The point is to surface execution problems early, while there is still time to fix them, not to rehearse a finished result. So that the demo is reproducible in the room, the implementation steward makes sure the team's current code and data state is pushed to GitHub before each of these two sessions.

The two presentations are graded together, on the seriousness of the preparation and on how usefully they expose the team's real implementation state and questions — **not on polish or on how good the interim results look**. Honest presentation of what is not yet working counts in the team's favor.

Grading (10 points): 5 for process and 5 for the in-class workshopping: the discussion the demo generates, not the polish of the presentation.

Assignment 5: Final team report

Each team submits one final report at the end of the standard exam period, roughly 10–15 pages plus appendices. The report must include the registered PAP, a "deviations and surprises" section, the analysis carried out following the PAP, any exploratory analysis clearly labeled as such, limitations, and an honest assessment of what the study does and does not support. All code and data necessary to reproduce the analysis must be supplied alongside the report; the report itself should be a literate document (Quarto, Rmd, Jupyter, or equivalent) so that results can be reproduced from the source.

The final report is the largest single component of the course grade and is graded on the credibility of the evidence the team produces, the honesty and clarity of the reporting (including how deviations and surprises are handled), and the quality of the writing, code, and figures. The report steward coordinates and submits the written report; the implementation steward is organizes the reproducible repository (code, data, and the evaluation pipeline) that accompanies it: making sure it is complete and runs.

In addition to the team report, **each student** submits a one-page reflection on their own experience of the project: what they contributed to the project, what they learned about evaluation methodology, what their team would do differently given what they now know, what group dynamics were difficult, what surprised them, and where they grew. The reflection is graded individually and is a chance to reflect on the work in a way that is not constrained by the team's collective voice.

Grading (30 points): 5 for process, 5 for the literature review, 5 for methods, 5 for the analysis, 5 for reproducibility, and 5 for the report overall. More detailed rubric will be provided in the course GitHub.

A detailed rubric will be provided in the course GitHub organization.

Participation

The participation grade assumes that students take part as active participants in the exchange, production, and critique of ideas (their own ideas and the ideas of others). Students should come to class having done the assigned reading and prepared to discuss it, and should also read their teammates' reflection memos before class. Participation is marked by its active nature, its consistency, and its quality. To receive the best possible participation grade, students should ask thoughtful questions, build on their colleagues' comments, and come prepared with specific observations about the readings and about the design of their own team's evaluation.

This class is structured as a workshop seminar in which student teams iterate on a single project across the whole term, so substantial participation is expected. Active reading (highlighting, taking notes, jotting down questions) will help students contribute well in class.

The participation grade covers in-class participation only.

Course policies:

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance: Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. Students do not need to inform the course instructor(s) in advance if they are unable to attend a session. For further information please consult the Examination Rules §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See Examination Rules §16, the Hertie Plagiarism Policy, and the Hertie Guidelines for Artificial Intelligence Tools.

Compensation for Disadvantages: If a student furnishes evidence that he or she is not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See Examination Rules §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

Project teams. Students will work in teams of approximately four. The class as a whole will pick the broad topics around which teams form (in week 2); within those topics, each team pursues its own evaluation question and its own design. Teams may pursue either a human-subjects experiment with AI as the treatment or a model-only evaluation grounded in a human task. The choice is up to each team.

Human subjects, ethics, and recruitment. Teams pursuing a human-subjects design will have access to a course-administered budget for online recruitment of participants. Ethics review will be arranged through Hertie; the timing and process will be discussed in class. Teams should plan recruitment carefully and aim to have data collection well underway in the first half of the implementation stage. Model-only teams do not recruit participants, but those that hire human raters or crowdworkers, or that use personal or scraped data, face the same ethics and data-protection obligations and complete the relevant sections of Assignment 2.

Compute and API access. Each student will be allocated a baseline of API credits at the start of the course. Beyond that baseline, additional resources can be requested as part of the team's pre-analysis plan, and must be justified by the design. Teams are free to choose their own evaluation framework and infrastructure, and should justify the choice in the PAP.

Code and reproducibility. All project artifacts (pre-analysis plans, evaluation code, analysis scripts, and final reports) should be maintained in a version-controlled repository. Reports should be literate (executable notebooks or equivalent) so that evaluation results can be reproduced from the code.

Working in the open on GitHub. Teams are expected to do their project work transparently on GitHub in a shared repository within the course organization. Work should happen *in the open* as it would on a real collaborative project: changes land through **pull requests**, not direct commits to the main branch, tasks and decisions are tracked as **issues**, and the team uses GitHub's **project tracking** (a board or equivalent) to organize and assign work. The instructor will look at the full GitHub history (commits, pull requests, reviews, issues, and the project board) when assessing the **process** component of Assignments 2–5 — that is, how well the team ran these practices, and how visibly and reliably each student stewarded their own piece and contributed to the parts stewarded by others. The goal is to make the team's real collaboration legible, not to chase activity metrics. A healthy repository shows distributed, well-attributed contribution across the whole term, not work that lands as a last-minute burst from one or two members. Students need not contribute equally on every dimension of the project. They must, however, make sure their own work is visible and legible through GitHub.

4. General Readings

This course has no single required textbook. The required readings are papers, book chapters, and online resources, listed per session in the next section and posted on Github. The works below are **reference textbooks** worth having at hand across the term: specific chapters are assigned in the relevant sessions, and the books are listed here as durable references.

- Gerber, A. S., & Green, D. P. (2012). *Field Experiments: Design, Analysis, and Interpretation*. W. W. Norton. *The experimental-design backbone of the course: randomization, estimation, and interpretation of field experiments. Ch. 1–4 are the core.*
- Japkowicz, N., & Boukouvalas, Z. (2024). *Machine Learning Evaluation: Towards Reliable and Responsible AI*. Cambridge University Press. *A full graduate text on evaluating ML systems: performance metrics, resampling/cross-validation, and which statistical test fits which model comparison. The reference for rigorous model comparison.*
- Sandvig, C. et al. (2026). [*Auditing AI*](#). MIT Press (Essential Knowledge series), open access. *A non-specialist guide to behavioural evaluation of consequential AI systems: what an audit is, how to run one, how to read someone else's, and what a healthy auditing ecosystem looks like.*
- Huyen, C. (2025). *AI Engineering: Building Applications with Foundation Models*. O'Reilly. *Practitioner's perspective on evaluation in product contexts; Ch. 4 covers evaluation.*
- Jurafsky, D., & Martin, J. H. [*Speech and Language Processing*](#) (3rd ed., draft). *Standard, freely available NLP reference. Useful self-contained sections on classification metrics and the paired-bootstrap significance test (§4.9–4.11) and on LLM evaluation, data contamination, and fairness (§7.6).*

5. Course Sessions and Readings

The course runs for twelve sessions of 100–110 minutes each. Most sessions open with **workshop time**, in which teams iterate on the milestone they developed over the past week, together with the instructor and the rest of the class, and supported by that session's required readings; class then closes with a short **framing lecture** from the instructor that previews the *next* week's milestone. So each topic

is met three times, in order: the lecture (delivered a week ahead), then the reading (done before the session that workshops it), then the workshop itself. The session topics below are deliberately kept at the level of theme and weekly milestone. The exact activities and readings for each week will be confirmed as the term approaches. Each session lists a few **required readings** and, where useful, a short **optional readings** list; the readings go with that session's *workshop*, while the lecture that closes the session points ahead to the following week. The *Auditing AI* book (Sandvig et al., listed in §4) is referred to here as "Sandvig et al." with the relevant chapter. All readings are posted on Github. Assignment deadlines are flagged below at the session where they fall; the authoritative deadlines and full assignment descriptions are in §3. Reading-reflection memos (Assignment 1) are not flagged per session because they are rolling: each student's three memos are due on Github 24 hours before the session whose reading they cover.

Session 1: The landscape of AI evaluation

Lecture (no workshop this week; preparing for the session 2 pitches): What does it mean to evaluate an AI system, and why is it hard? This opening session is a survey of the contemporary toolkit for evaluating AI systems, and it sets up the central tension of the course: **benchmarks** (model-only evaluations run at scale against fixed tasks) versus **human experiments** (studies of whether a person actually does their task better with the AI). We walk through the main instruments students will meet and use: benchmarks and leaderboards, automated and programmatic scoring, **LLM-as-judge**, **human annotation**, and **task-based human-subjects experimentation**. For each, we look at where it is and isn't trustworthy. The running theme is that "which model is better" depends on what you ask, how you ask it, and how you score it; a *capability* question ("can the model do X?") differs from a *deployment* question ("does using it help someone do Y?"). The session also introduces the two design archetypes available in the course (a human-subjects experiment with AI tools as the intervention, or a model-only evaluation grounded in a real human task) and walks through how to sharpen an evaluation question: distinguishing the AI system from the task from the user, and separating capability questions from deployment questions. The aim is for students to leave able to begin forming their pitches for week 2.

Required readings:

- Wallach, H. et al. (2025). ["Position: Evaluating Generative AI Systems Is a Social Science Measurement Challenge."](#) arXiv:2502.00561.
- Huyen, C. (2025). *AI Engineering*, ch. 4, on the modern evaluation toolkit: benchmarks, LLM-as-judge, and comparative ranking.

Optional readings:

- Raji, I. D., et al. (2021). ["AI and the Everything in the Whole Wide World Benchmark."](#) NeurIPS Datasets and Benchmarks.
- Stanford CS, ["Evaluating LLMs" lecture](#), a survey of the major tools in this space.
- Recht, B. ["Demo or Die"](#), on demos versus real evaluation, sharpening the capability-vs-deployment distinction.

Session 2: Problem pitches and team formation

Workshop: Each student gives a short lightning talk (at most 3 slides, 5 minutes) pitching an evaluation question, in the form *a user, doing a task where an AI system might help, plus a reason to doubt that it does*. The question should be specific enough that a team could imagine running the study within a semester. Students vote on a small number of broad topic areas, and teams of approximately four form around them.

Lecture (previewing session 3): Measurement theory for AI evaluation — the move from background concept to systematized concept to instrument to measurement, and construct validity.

Required readings: one of each evaluation paradigm — a deployment experiment, a model-only benchmark, and a human-preference arena.

- Dell'Acqua, F. et al. (2023). ["Navigating the Jagged Technological Frontier."](#) Harvard Business School Working Paper. A human-subjects experiment measuring whether consultants do their task better with AI.
- Hendrycks, D. et al. (2021). ["Measuring Massive Multitask Language Understanding."](#) ICLR. MMLU, a very high-profile model-only benchmark with no humans in the loop.
- Chiang, W.-L. et al. (2024). ["Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference."](#) ICML. Comparative ranking from crowdsourced pairwise human preferences (LMArena).

Optional readings:

- Brynjolfsson, E., Li, D., & Raymond, L. (2025). ["Generative AI at Work."](#) *OJE*.
- METR (2025). ["Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity."](#)
- Liao, Q. V., & Xiao, Z. (2023). ["Rethinking Model Evaluation as Narrowing the Socio-Technical Gap."](#)
- Recht, B. ["Rossi's Metallic Rules"](#), on Rossi's iron law of evaluation and why interventions so often fail to show an effect.

A catalog of examples (browse — don't read all): Skim this list to see the range of what "an evaluation" can be; pick a few that resonate with your pitch. Grouped by paradigm.

Model-only benchmarks (capability):

- Wang, A. et al. (2019). ["GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding."](#) ICLR. Nine-task NLU aggregate that became the standard pre-/early-BERT leaderboard.
- Wang, A. et al. (2019). ["SuperGLUE: A Stickier Benchmark for General-Purpose Language Understanding Systems."](#) NeurIPS. Harder successor to GLUE once models hit near-human scores.
- Chen, M. et al. (2021). ["Evaluating Large Language Models Trained on Code."](#) arXiv. Introduces HumanEval: functional correctness of generated Python via unit tests.
- Jimenez, C. E. et al. (2023). ["SWE-bench: Can Language Models Resolve Real-World GitHub Issues?"](#) ICLR. Models edit real repos to fix genuine GitHub issues, graded by the projects' own tests.
- Rein, D. et al. (2023). ["GPOA: A Graduate-Level Google-Proof Q&A Benchmark."](#) arXiv. Expert-written science questions that stay hard even with web access.
- Srivastava, A. et al. (2022). ["Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models."](#) TMLR. The 200+-task collaborative BIG-bench suite.
- Cobbe, K. et al. (2021). ["Training Verifiers to Solve Math Word Problems."](#) arXiv. Introduces GSM8K, the grade-school multi-step arithmetic benchmark.
- Deng, J. et al. (2009). ["ImageNet: A Large-Scale Hierarchical Image Database."](#) CVPR. The vision benchmark whose annual challenge drove the deep-learning era.
- Lin, S. et al. (2021). ["TruthfulQA: Measuring How Models Mimic Human Falsehoods."](#) ACL. Tests whether models reproduce common human misconceptions.

- Phan, L. et al. (2025). ["Humanity's Last Exam."](#) arXiv. Frontier expert-level benchmark built to restore headroom after MMLU saturation.

Human-subjects deployment experiments (does the AI actually help?):

- Peng, S. et al. (2023). ["The Impact of AI on Developer Productivity: Evidence from GitHub Copilot."](#) arXiv. RCT: developers building an HTTP server finished 55.8% faster with Copilot.
- Noy, S., & Zhang, W. (2023). ["Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence."](#) Science. Preregistered writing-task experiment; ChatGPT cut time ~40% and raised quality.
- Bansal, G. et al. (2021). ["Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance."](#) CHI. Whether AI advice + explanations beat human-alone or AI-alone accuracy.
- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). ["To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making."](#) CSCW. Interventions that reduce overreliance on wrong AI advice.
- Green, B., & Chen, Y. (2019). ["The Principles and Limits of Algorithm-in-the-Loop Decision Making."](#) CSCW. How showing algorithmic predictions changes human risk-assessment decisions.
- Vaccaro, M., Almaatouq, A., & Malone, T. (2024). ["When Combinations of Humans and AI Are Useful: A Systematic Review and Meta-Analysis."](#) Nature Human Behaviour. 106 studies on whether human-AI teams beat the best of either alone.
- Tschandi, P. et al. (2020). ["Human-Computer Collaboration for Skin Cancer Recognition."](#) Nature Medicine. Clinicians diagnosing skin lesions with vs. without AI support.

Human-preference arenas & LLM-as-judge:

- Chiang, W.-L. et al. (2024). ["Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference."](#) ICML. The platform paper: ranks models from crowdsourced **human** pairwise preferences (Elo). (*Also the required reading above — shown here to contrast with its LLM-as-judge sibling below.*)
- Zheng, L. et al. (2023). ["Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena."](#) NeurIPS. Introduces MT-Bench and **LLM-as-judge** (GPT-4 as automatic evaluator), validated against the arena's human votes; documents judge biases.
- Dubois, Y. et al. (2024). ["Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators."](#) COLM 2024. An off-the-shelf LLM-judge win-rate evaluator that regresses out length bias.
- Köpf, A. et al. (2023). ["OpenAssistant Conversations — Democratizing Large Language Model Alignment."](#) NeurIPS. A large crowdsourced human-annotated preference/conversation corpus.
- Bai, Y. et al. (2022). ["Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback."](#) arXiv. Canonical human pairwise-preference data collection for evaluation/training.

Safety & dangerous-capability evaluations:

- Shevlane, T. et al. (2023). ["Model Evaluation for Extreme Risks."](#) arXiv. Framework pairing dangerous-capability with alignment evaluations.
- Phuong, M. et al. (2024). ["Evaluating Frontier Models for Dangerous Capabilities."](#) arXiv. DeepMind battery probing persuasion, cyber, and self-proliferation on Gemini.

- Greenblatt, R. et al. (2024). ["AI Control: Improving Safety Despite Intentional Subversion."](#) ICML. Protocols keeping an untrusted model safe even when it tries to subvert oversight.
- Mazeika, M. et al. (2024). ["HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal."](#) ICML. Standardized red-teaming benchmark across attacks, models, and defenses.

Fairness & bias audits (classic / deployed systems):

- Bertrand, M., & Mullainathan, S. (2004). ["Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination."](#) American Economic Review. The canonical resume-correspondence audit of hiring discrimination.
- Angwin, J. et al. (2016). ["Machine Bias."](#) ProPublica. Audit of the COMPAS recidivism tool's racial error-rate disparities.
- Buolamwini, J., & Gebru, T. (2018). ["Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification."](#) PMLR. Commercial face-classification APIs with large accuracy gaps by skin tone and gender.
- Obermeyer, Z. et al. (2019). ["Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations."](#) Science. A deployed health-risk algorithm that underestimated Black patients' needs.

LLM bias & viewpoint audits:

- Westwood, S. J. et al. (2025). ["Measuring Perceived Slant in Large Language Models Through User Evaluations."](#) Stanford GSB Working Paper. 180k+ paired-output ratings from 10k U.S. respondents across 24 LLMs; nearly all are perceived as left-leaning, and a neutral-stance prompt measurably reduces it.
- Santurkar, S. et al. (2023). ["Whose Opinions Do Language Models Reflect?"](#) ICML. The OpinionQA audit comparing LM outputs to U.S. Pew survey responses, finding skew away from the general public.
- Durmus, E. et al. (2023). ["Towards Measuring the Representation of Subjective Global Opinions in Language Models."](#) arXiv. Anthropic's GlobalOpinionQA: default model opinions align most with the USA and parts of Europe/South America.
- Rozado, D. (2024). ["The Political Preferences of LLMs."](#) PLOS ONE. 11 political-orientation tests across 24 conversational LLMs (most diagnosed left-of-center; base models show no consistent slant).
- Motoki, F. et al. (2024). ["More Human than Human: Measuring ChatGPT Political Bias."](#) Public Choice. Impersonation-based audit finding ChatGPT's defaults favor U.S. Democrats, UK Labour, and Brazil's Workers' Party.

Session 3: Construct and problem setup

Workshop: Teams come to class having done a literature review on their topic, and workshop their construct and problem setup: *what is the downstream task, who is the user, what does "success" mean, and what have others measured and how?* By the end of class each team commits to a design archetype (human-subjects experiment or model-only evaluation) with an explicit argument for the choice.

Lecture (previewing session 4): Task design and elicitation — prompt sensitivity, scaffolding choices, and the elicitation-effort problem.

Required readings:

- Jacobs, A. Z., & Wallach, H. (2021). ["Measurement and Fairness."](#) FAccT.

- Adcock, R., & Collier, D. (2001). ["Measurement Validity: A Shared Standard for Qualitative and Quantitative Research."](#) APSR.
- Bean, A. M. et al. (2025). ["Measuring what Matters: Construct Validity in Large Language Model Benchmarks."](#) NeurIPS Datasets and Benchmarks.

Session 4: Instrument design (defining the intervention)

Workshop: The central question for the week: **what exactly is the treatment?** The "AI system" is more than a model: it is a model plus a prompting strategy plus a scaffold plus an interface plus, in human-subjects designs, an onboarding and a workflow. Teams operationalize all of this, and begin the ethics work that feeds Assignment 2 (the ethics and responsible-evaluation self-assessment): human-subjects teams the Hertie ethics-review process and recruitment, model-only teams their evaluation data and responsibility statement.

Lecture (previewing session 5): Scorers and their validation — exact-match, reference-based, LLM-as-judge, and human rating, plus inter-rater reliability.

Required readings:

- Ribeiro, M. T. et al. (2020). ["Beyond Accuracy: Behavioral Testing of NLP Models with CheckList."](#) ACL. A worked methodology for designing a test instrument rather than relying on aggregate accuracy.
- Lai, V. et al. (2023). ["Towards a Science of Human-AI Decision Making: An Overview of Design Space in Empirical Human-Subject Studies."](#) FAccT. A survey of how human-AI studies operationalize the intervention (task, model, and assistance elements such as explanations, confidence, and onboarding) — the human-subjects counterpart to CheckList.

Optional readings:

- Sandvig et al., *Auditing AI*, ch. 3 ("The Steps of an AI Audit"), a worked example of operationalizing an evaluation design.
- Apollo Research (2024). ["We Need a Science of Evals."](#)
- Husain, H. ["Your AI Product Needs Evals."](#) hamel.dev.

Session 5: Outcome measurement and scorer validation

Workshop: Teams workshop their outcome measurement strategy: *what specifically will be measured, and how will responses be scored?* If using human raters, how are they trained and calibrated? If using an LLM-as-judge, how is the judge validated against human judgment? For human-subjects teams, this is also where outcomes beyond raw task performance enter: appropriate reliance, over-reliance, confidence calibration.

Lecture (previewing session 6): Experimental design for AI evaluation — variance components (person, task, prompt, seed, model), power with multiple sources of variance, and contamination as a canonical threat.

Required readings:

- Pangakis, N., Wolken, S., & Fasching, N. (2023). ["Automated Annotation with Generative AI Requires Validation."](#)
- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). ["To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI."](#) CSCW.

Optional readings:

- Chehbouni, K. et al. (2025). ["Neither Valid nor Reliable? Investigating the Use of LLMs as Judges."](#)
- Zheng, L. et al. (2023). ["Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena."](#)

Session 6: Experimental design (randomization and threats to validity)

Due before this session: Assignment 2 — Ethics and responsible-evaluation self-assessment (§3).

Workshop: Teams present their experimental design: *what is randomized, at what level, with what blocking or stratification, and what are the threats to validity specific to this study?* The issues differ by archetype: randomization, within- vs. between-subjects, and contamination for human-subjects teams; item sampling, prompt variation, and benchmark contamination for model-only teams.

Lecture (previewing session 7): Analysis for AI evaluation — covariate adjustment in human-AI studies, AI-specific complications (refusals, drift, model updates mid-experiment), and what a credible eval report looks like.

Required readings:

- Sandvig et al., *Auditing AI*, ch. 4 ("Interpreting Audit Results"), focusing on the *framing* sections.
- Sainz, O. et al. (2023). ["NLP Evaluation in Trouble: On the Need to Measure LLM Data Contamination for Each Benchmark."](#)
- Gerber, A. S., & Green, D. P. (2012). *Field Experiments*, ch. 3–4.

Optional readings:

- Miller, E. (2024). ["Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations."](#)
- Recht, B. ["The Adaptivity Paradox"](#), on test-set reuse and leaderboard overfitting, complementing the Sainz reading on contamination.

Session 7: Analysis plan

Workshop: Each team's analysis plan: *primary and secondary hypotheses; estimation procedure; covariate adjustment; handling of missing data and non-compliance (participants who don't engage with the AI, AI refusals, hallucinations); multiple-comparisons correction; defenses against contamination and elicitation concerns.*

Lecture (previewing session 8): What makes a credible pre-analysis plan — statistical power with multiple sources of variance, and standard operating procedures as a safety net for the decisions a PAP must settle in advance.

Required readings:

- Lin, W. (2013). ["Agnostic Notes on Regression Adjustments to Experimental Data."](#) *Annals of Applied Statistics*.
- Liang, P. et al. (2022). ["Holistic Evaluation of Language Models"](#) (HELM), read selectively.

Optional readings:

- Athey, S., & Imbens, G. W. (2017). ["The Econometrics of Randomized Experiments."](#) *Handbook of Economic Field Experiments*.

Session 8: Pre-analysis plan workshop and sign-off

Due before this session: Assignment 3 — Pre-analysis plan, registered to OSF (§3).

Workshop: Teams submit a complete, literate pre-analysis plan in advance of class. Class time is spent on **structured peer review**: each team reads another team's PAP in depth and stress-tests it, with written instructor feedback provided beforehand. By the end of the day, PAPs are **final and registered** to OSF or an equivalent public registry; from this point any deviation must be documented and justified in the final report. The remaining sessions shift to implementation.

Lecture (previewing session 9): Building and running an evaluation — turning the registered design into a working pipeline — and the common implementation failure modes (rate limits, API errors, participant dropout, prompt-format issues, scorer disagreements, cost overruns) and how to triage and document deviations from the PAP.

Required readings:

- Coppock, A. (2020). ["10 Things to Know About Statistical Power."](#) EGAP Resource.
- Lin, W., & Green, D. P. (2016). ["Standard Operating Procedures: A Safety Net for Pre-Analysis Plans."](#) *PS: Political Science & Politics*, 49(3), 495–500, together with the living [Green Lab Standard Operating Procedures](#) document.

Session 9: Implementation I (execution kickoff and pilot results)

Assessed this session: Assignment 4 — Implementation work-in-progress presentation (§3).

Workshop: Teams begin running their evaluations during the week between sessions 8 and 9, and present pilot results in workshop format: the first small-N run, what worked, what is already surprising. This is the first of the two implementation work-in-progress presentations run by the implementation steward (Assignment 4): a deliberately informal code-demo and status report — current pipeline state, first results, what is breaking, and the specific questions the team most wants feedback on. This is deliberately early so that design issues that surface only in execution can be caught while there is still time to respond. The theme of the week is operational: a design that is clean on paper still has to *run*, and the point of a pilot is to find out where it doesn't before the real sample is spent.

Lecture (previewing session 10): Honest reporting — the difference between pre-registered hypotheses, exploratory analysis, and post-hoc rationalization, the researcher degrees of freedom that turn an innocent-looking analysis into a foregone conclusion, and how to write up a study where things did not go as planned.

Reading is deliberately light this week — teams should be running experiments.

Required readings:

- Karlan, D., & Appel, J. (2016). *Failing in the Field: What We Can Learn When Field Research Goes Wrong*. Princeton University Press. Selected chapters on how field studies break in execution and why piloting is the cheapest insurance against it.

Optional readings:

- Sandvig et al., *Auditing AI*, ch. 4, *Why Audits Go Wrong* section. The AI-evaluation counterpart: how audits fail in execution rather than in conception.
- Selected "things went wrong" sections from published evaluation reports (linked on Github).

Session 10: Implementation II (troubleshooting and mid-course corrections)

Assessed this session: Assignment 4 — Implementation work-in-progress presentation (§3).

Workshop: Teams present current status: data collected, issues encountered, proposed adjustments. This is the second implementation work-in-progress presentation run by the implementation steward

(Assignment 4), in the same informal code-demo and status format as session 9. The class collectively troubleshoots, and any proposed deviations from the PAP are discussed openly and documented — informed by last week's lecture on honest reporting and the garden of forking paths. Where session 9 was about whether the study *runs*, this week is about catching the problems while they can still be fixed.

Lecture (previewing session 11): Analysis, uncertainty, and communicating results — estimation and covariate adjustment, error bars on evals, multiple-comparisons correction, communicating uncertainty to non-technical stakeholders, and what claims the evidence actually supports.

Required readings:

- Gelman, A., & Loken, E. (2014). ["The Statistical Crisis in Science."](#) *American Scientist*, 102(6), 460–465. The "garden of forking paths": how analytic flexibility inflates false positives even without any intent to p-hack — the case for committing to a PAP and labeling everything else exploratory.
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). ["False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant."](#) *Psychological Science*, 22(11). The canonical demonstration of researcher degrees of freedom, with concrete disclosure rules.

Session 11: Analysis and draft findings

Workshop: Teams present preliminary analysis following their PAP: *what do the results look like, where is there uncertainty, and what secondary or exploratory analyses are worth running (labeled as such)?*

Lecture (previewing session 12): Reproducibility and release — packaging code and data so a third party can re-run the evaluation, environment and version pinning, literate executable reports, evaluation cards and datasheets for the eval artifact, and documenting deviations from the PAP for release. This prepares the final presentations and the reproducible report and repository that follow.

Required readings:

- Miller, E. (2024). ["Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations."](#)
- Gelman, A., Hill, J., & Yajima, M. (2012). ["Why We \(Usually\) Don't Have to Worry About Multiple Comparisons."](#) *Journal of Research on Educational Effectiveness*, 5(2).

Optional readings (for the reproducibility and release lecture):

- Gebru, T. et al. (2021). ["Datasheets for Datasets."](#) *Communications of the ACM*.
- Mitchell, M. et al. (2019). ["Model Cards for Model Reporting."](#) FAccT.
- Biderman, S. et al. (2024). ["Lessons from the Trenches on Reproducible Evaluation of Language Models."](#) The reproducibility design behind the lm-eval-harness.
- Kapoor, S., & Narayanan, A. (2023). ["Leakage and the Reproducibility Crisis in Machine-Learning-Based Science."](#) *Patterns*.

Session 12: Final presentations

Workshop: Each team gives a final 20–25 minute presentation: question, design, what happened, what the results support, what the limitations are, and what they would do differently. Other teams and the instructor provide final feedback before written reports are submitted.

Optional readings:

- Sandvig et al., *Auditing AI*, ch. 5 ("After the Audit") and ch. 6 ("A Healthy Audit Ecosystem"), optional further reading for students whose interest turns to the consequences of evaluation.

After session 12: final written report

Due: Assignment 5 — Final team report and individual one-page reflection, end of the standard exam period (§3).

Final written reports are due at the end of the standard exam period, together with each student's individual one-page process reflection. During the exam period the instructor will hold office hours for students to discuss their projects, potential follow-on work, and evaluation ideas relevant to theses or future research.